EXPERIMENT - 4 DATE: 10/08/26

AIM: To design, set up and configure a LAN connecting four computers through a network switch, assign static IP address to all devices, verify connectivity using the ping command, and demonstate resource/data sharing accross the network

Procedure:

Step 1: From device catagory panel, select End Devices and drag 4 PCs onto the workspace. Select switches and drag one onto workspace.

Step 2: Cable the Device

Click Connection → Copper - Straight Through Cable.
Click on PC select available Fast Ethernet
Repeat for all 4 Ports.

Step 3: Assign IP Address to PCs

Click PC → Desktop tab → IP Configuration.
Enter Static IP Address and Subnet Mask.

Step 4: Configure the Switch Management IP

Click Switch → CLI tab

Enter the following:

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

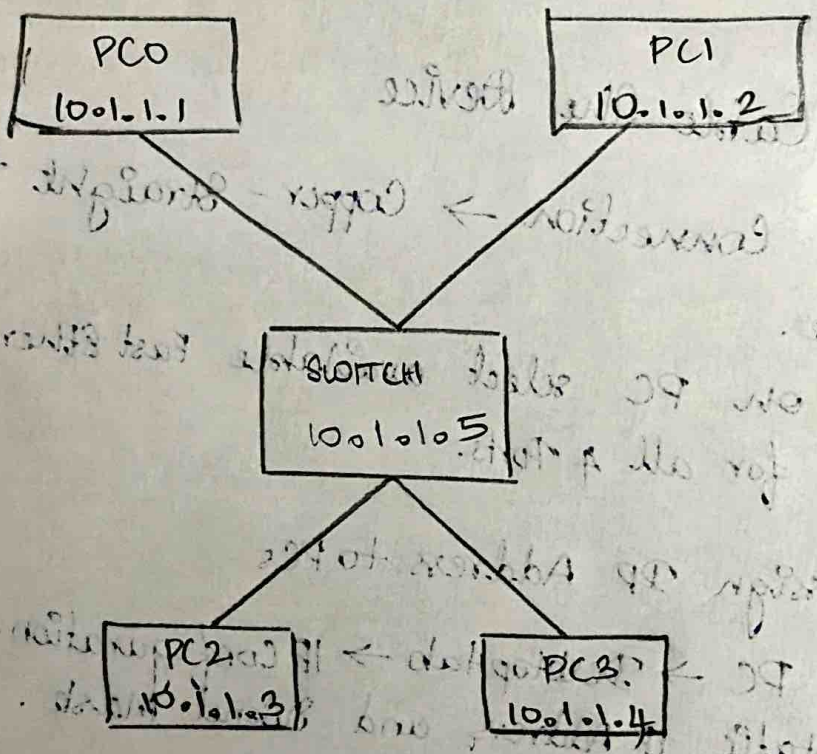
Step 5: Verify Connectivity Using ping

Click on PC → Desktop tab → Command Prompt.

Step 6: Demonstrate Data Sharing

Add a server

Step 7: Observe Packet Flow in Simulation Mode



RESULT: Thus, all four PCs were successfully connected. Shared folders was successfully accessed from other computers.